



Airlink Freeradius Billing System V3.0

PREPARED FOR

Nepal Airlink

PREPARED BY

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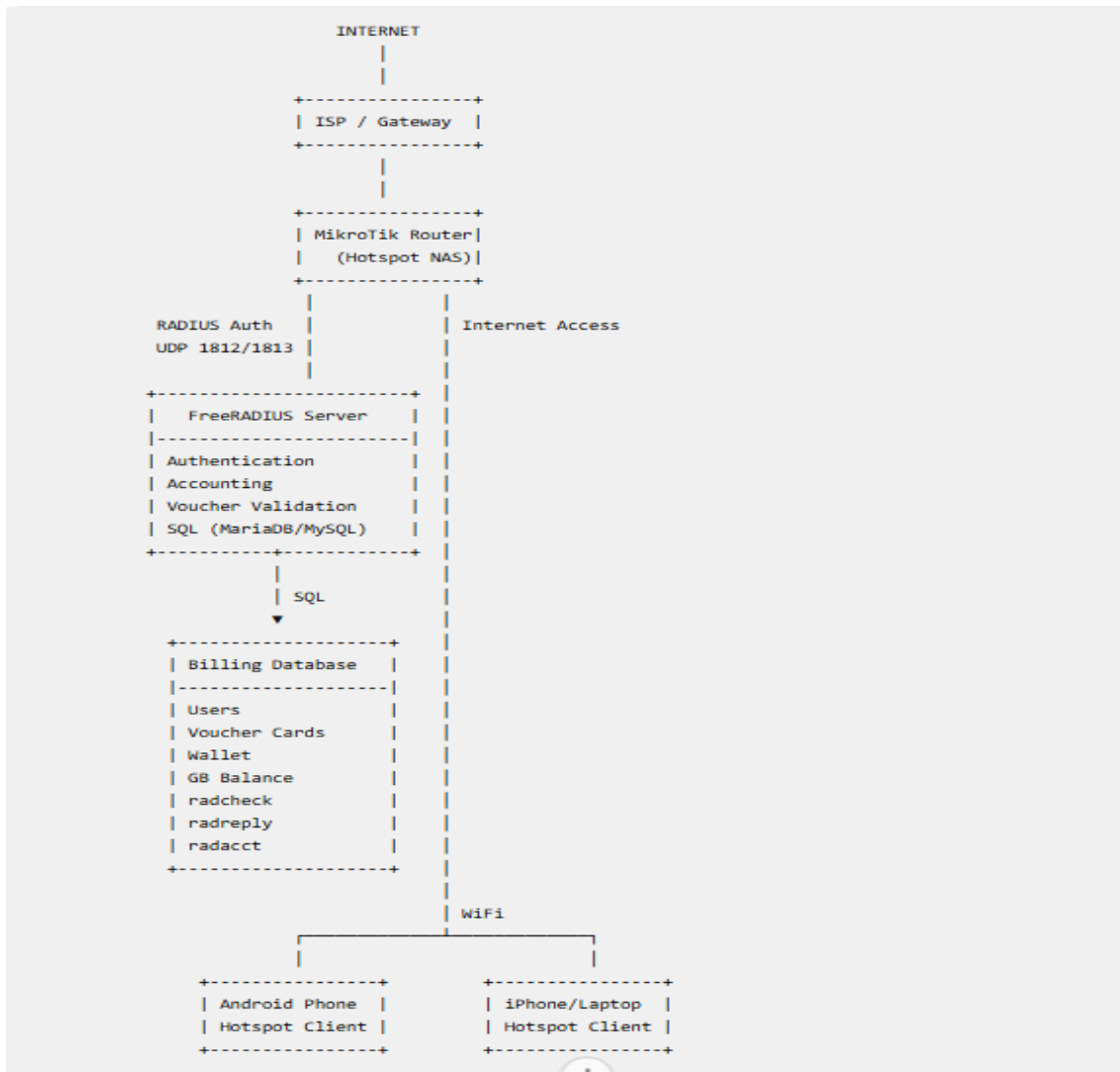
1. Project Overview

The Nepal Airlink Billing System Version 3.0 is a major upgrade of the existing Version 2.0 platform designed to improve operational efficiency, scalability, and business expansion. The upgraded system introduces a multi-level distribution model that enables resellers to appoint and manage seller parties for the distribution of internet packages on a GB-based billing model.

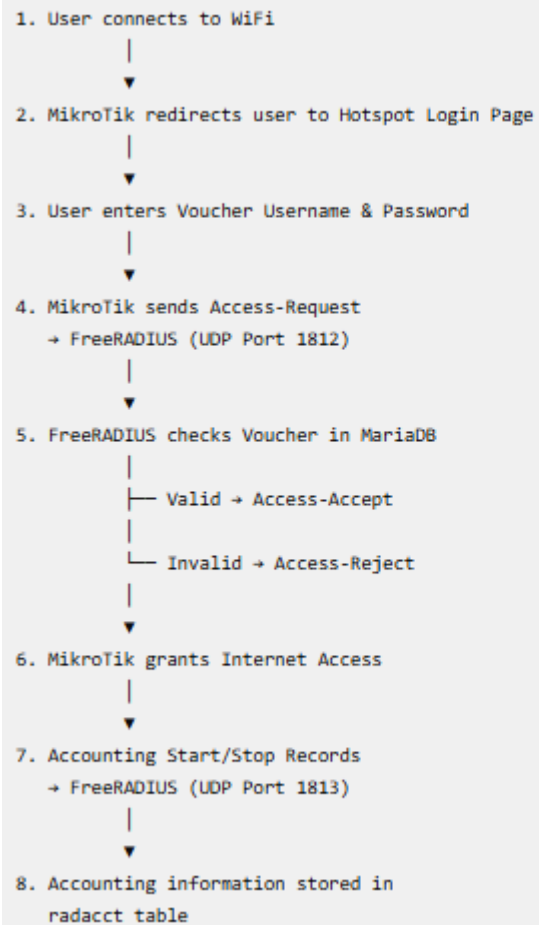
The new version supports modern ISP business requirements, allowing Nepal Airlink to expand its market reach in remote and tourism-focused regions by empowering local businesses such as hotels, lodges, travel agencies, and retail shops to sell internet packages directly to end customers.

A. Current Running System v2.0 Feature

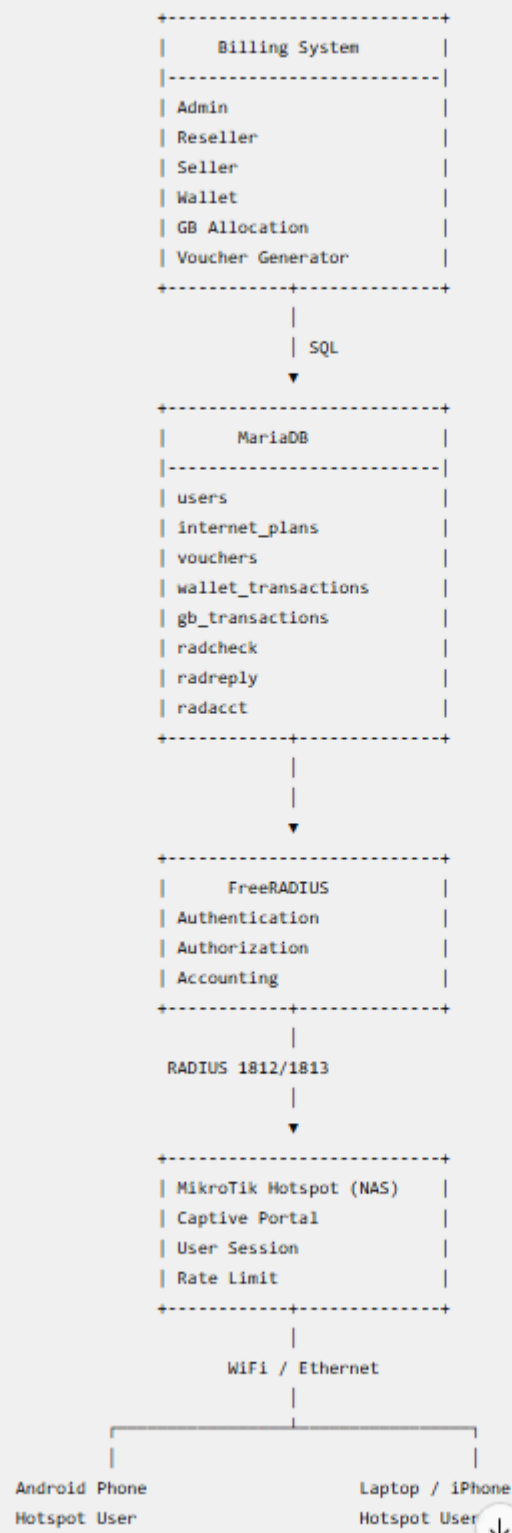
Below is a simple network architecture diagram showing how **Mobile Clients**, **MikroTik Hotspot (NAS)**, and **FreeRADIUS Server** interact in a hotspot authentication environment.



Authentication Flow



Complete System Architecture



B. UI for the 2.0 Admin user

Voucher Card Management Module

1. Overview

The Voucher Card Management Module is designed to enable the Administrator and Reseller to generate, distribute, monitor, and manage prepaid Internet voucher cards. The system supports both regular voucher generation for day-to-day sales and batch voucher generation for mass production. Role-based access control ensures that each user can only access voucher information according to their assigned permissions.

2. Administrator Functions

The Administrator has complete access to the voucher management system and can generate vouchers for direct retail sales as well as on behalf of any reseller.

2.1 Regular Voucher Generation

Regular Voucher Generation is intended for small-scale sales where a limited number of voucher cards are required.

The administrator can:

- Select Internet Package.
- Select Voucher Validity.
- Select Reseller (optional).
- Enter Quantity (e.g., 10, 20, 50, or 100 cards).
- Generate voucher cards instantly.
- Export vouchers as printable PNG images.
- Print voucher cards directly.
- Download individual voucher images.

Each voucher card shall contain:

- Voucher Code
- Username
- Password (if applicable)
- Package Name
- Data (GB) or Time Limit
- Validity Period
- QR Code
- Company Logo
- Terms and Conditions



2.2 Batch Voucher Generation

Batch Voucher Generation is designed for large-scale voucher production.

Typical quantities include:

- 500 Cards
- 1,000 Cards
- 2,000 Cards
- 5,000 Cards
- 10,000 Cards
- 20,000 Cards
- Custom Quantity

The Administrator shall be able to:

- Select Package
- Select Reseller
- Enter Quantity
- Generate Voucher Series
- Export Complete Voucher List to Microsoft Excel (.xlsx)
- Export Voucher Codes as CSV
- Download Batch Report
- Print Voucher Sheets

Each generated batch shall receive a unique Batch ID.

Example:

Batch ID	Package	Quantity	Generated By	Date
BAT240701	5 GB	1000	Admin	2026-07-07

3. Voucher Ownership

Every generated voucher shall have an owner.

Example:

Voucher	Owner
VC000001	Admin
VC000002	Reseller A
VC000003	Seller X

The ownership field ensures complete voucher tracking.

4. Used Voucher Report

The Administrator can view all voucher usage across the system.

Dashboard Example

Used Cards

Total Used

12,560

When the Administrator clicks **Used Cards**, the system shall display a package summary.

Example

Package	Generated	Used	Remaining
1 GB	10,000	8,000	2,000
5 GB	5,000	4,200	800
10 GB	2,000	1,250	750

Package Detail

When the Administrator clicks on the **Used** value of a package, the system displays all vouchers that have been consumed.

Example

Package: 5 GB

Voucher Code	User	Login Date	Reseller	Seller
VC100001	hotel01	2026-07-07	Everest ISP	Namche Shop
VC100002	lodge02	2026-07-07	Everest ISP	Lukla Store
VC100003	user125	2026-07-07	Everest ISP	Phakding Shop

Filters shall include:

- Date Range
- Package
- Reseller
- Seller
- Voucher Code
- Username
- Status

5. Voucher Status

Each voucher shall have one of the following statuses:

Status	Description
New	Voucher generated but not sold
Sold	Voucher sold but not yet used
Active	Voucher currently in use
Expired	Validity period ended
Disabled	Voucher blocked by administrator

6. Reseller Functions

A Reseller can generate voucher cards only for their own business.

The Reseller shall be able to:

- Generate Regular Voucher Cards.
- Generate Batch Voucher Cards.
- Download PNG Voucher Cards.
- Export Voucher List to Excel.
- Print Voucher Sheets.
- View Voucher History.
- Search Voucher by Code.
- View Voucher Status.
- View Used Voucher Reports.

Restrictions:

- Cannot view vouchers belonging to other resellers.
- Cannot edit vouchers created by another reseller.
- Cannot delete administrator vouchers.
- Cannot access system-wide sales reports.

7. Reseller Used Voucher Report

The Reseller dashboard displays only their own voucher usage.

Example

Package	Generated	Used	Remaining
5 GB	1000	750	250
10 GB	500	420	80

Clicking the **Used** count opens a detailed list showing only vouchers owned by that reseller.

Example

Voucher Code	Customer	Login Date	Status
VC500001	User001	2026-07-07	Used
VC500002	User002	2026-07-07	Used

ADMINISTRATOR

ADMINISTRATOR

HOME

CUSTOMER

PLAN

HOTSPOT

NAS

MONITOR NAS

WALLET

ADMINISTRATION

Total Voucher

6022

View Reports

Active Hotspot

0

View Reports

Used/Expired Voucher

727

View Reports

Damaged/Returned Voucher

0000

View Reports

Total PPPOE Users

19

View Reports

Active PPPOE

0

View Reports

Expired User

0000

View Reports

Disabled User

0000

View Reports

POS USER

Seller	Voucher	Used	Stock	Damage/Return
migmakofhe	2020	395	1625	0
Upstaredi	1999	113	1886	0
Anggita	1997	218	1779	0
adris	6	1	5	0

VOUCHER ALLOCATION SUMMARY

Allocate Voucher

E.N.

Voucher Collector

Total Voucher

Used Voucher

Unused Voucher

Id Start At

Id End At

NEPAL AIRLINK
Search Customers... 🔍
🔑 Login

HOME

 CUSTOMER

 PLAN

- Hotspot Plans
- Service Plans
- Bandwidth Plans

HOTSPOT PLANS

+ Add Hotspot Plan
+ Edit
+ Delete

S.N.	Plan Name	Plan Type	Bandwidth	Data Limit	Plan Validity	Plan Price	Time Limit	API NAS
☐	24Hour_Pack	Unlimited	5Mbps	0 MB	1 d	1800	0 Hrs	0
☐	48Hour_Pack	Unlimited	5Mbps	0 MB	2 d	1200	0 Hrs	0
☐	FDC_15Days_Package	Unlimited	2Mbps	0 MB	15 d	0	0 Hrs	0
☐	12Hour_Pack	Unlimited	10Mbps	0 MB	12 h	1200	0 Hrs	0
☐	local_24Hour_Pack	Unlimited	7Mbps	0 MB	1 d	200	0 Hrs	0
☐	Monthly_staff_package	Unlimited	2Mbps	0 MB	30 d	1500	0 Hrs	0

1
Next
Last

HOME

 CUSTOMER

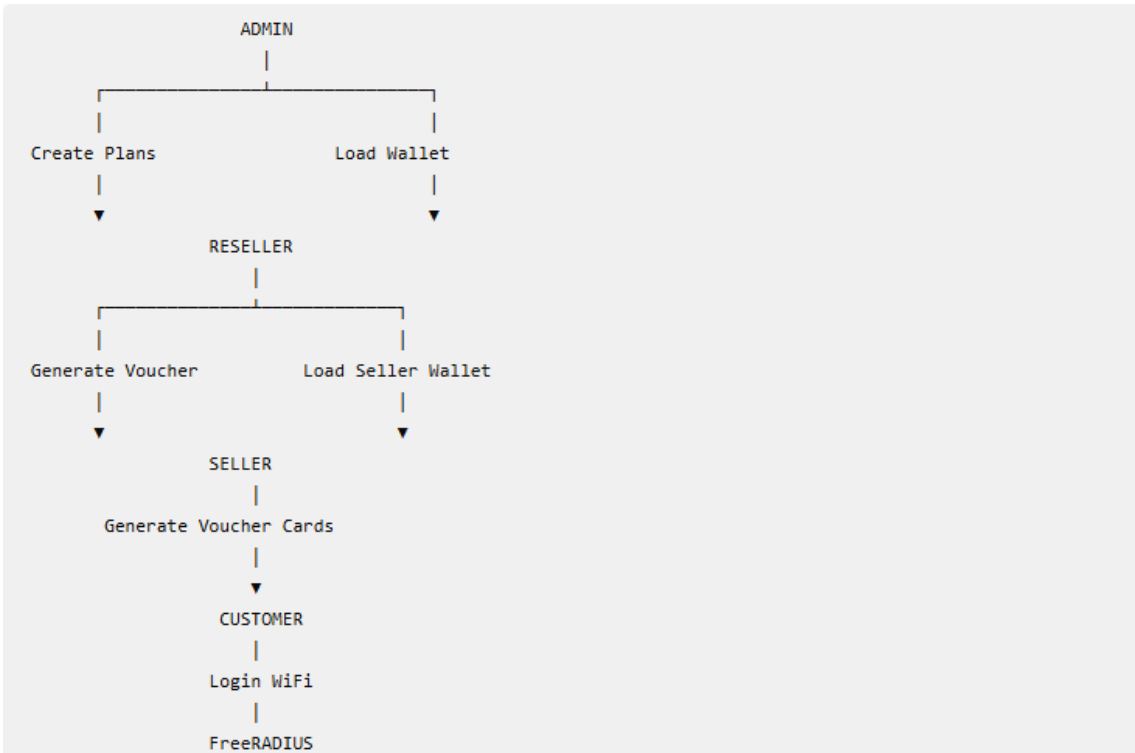
- User Details
- Search Details
- BPOCE Users
- Settings

MANAGE CONTACT

+ Add New
+ Change Password
+ Disable
+ Activate
+ Deactivate
+ Recharge
+ Charge(MM)

S.N.	Username	Profile	Batch	Created_date	POS Owner
☐	2AAAEI9	Monthly_staff_package		2026-06-15 16:35:55	admin
☐	C24E78	Monthly_staff_package		2026-06-15 16:35:55	admin
☐	E76144	local_24Hour_Pack	local	2026-06-10 17:07:07	admin
☐	9A4E14	local_24Hour_Pack	local	2026-06-10 17:07:07	admin
☐	A03D76	local_24Hour_Pack	local	2026-06-10 17:07:07	admin
☐	825DFA	local_24Hour_Pack	local	2026-06-10 17:07:07	admin
☐	B7EEED	local_24Hour_Pack	local	2026-06-10 17:07:07	admin
☐	96388D	local_24Hour_Pack	local	2026-06-10 17:07:07	admin
☐	49B3C5	local_24Hour_Pack	local	2026-06-10 17:07:07	admin
☐	259F0D	local_24Hour_Pack	local	2026-06-10 17:07:07	admin

2. Multi-Level Wallet & Voucher Distribution System 3.0

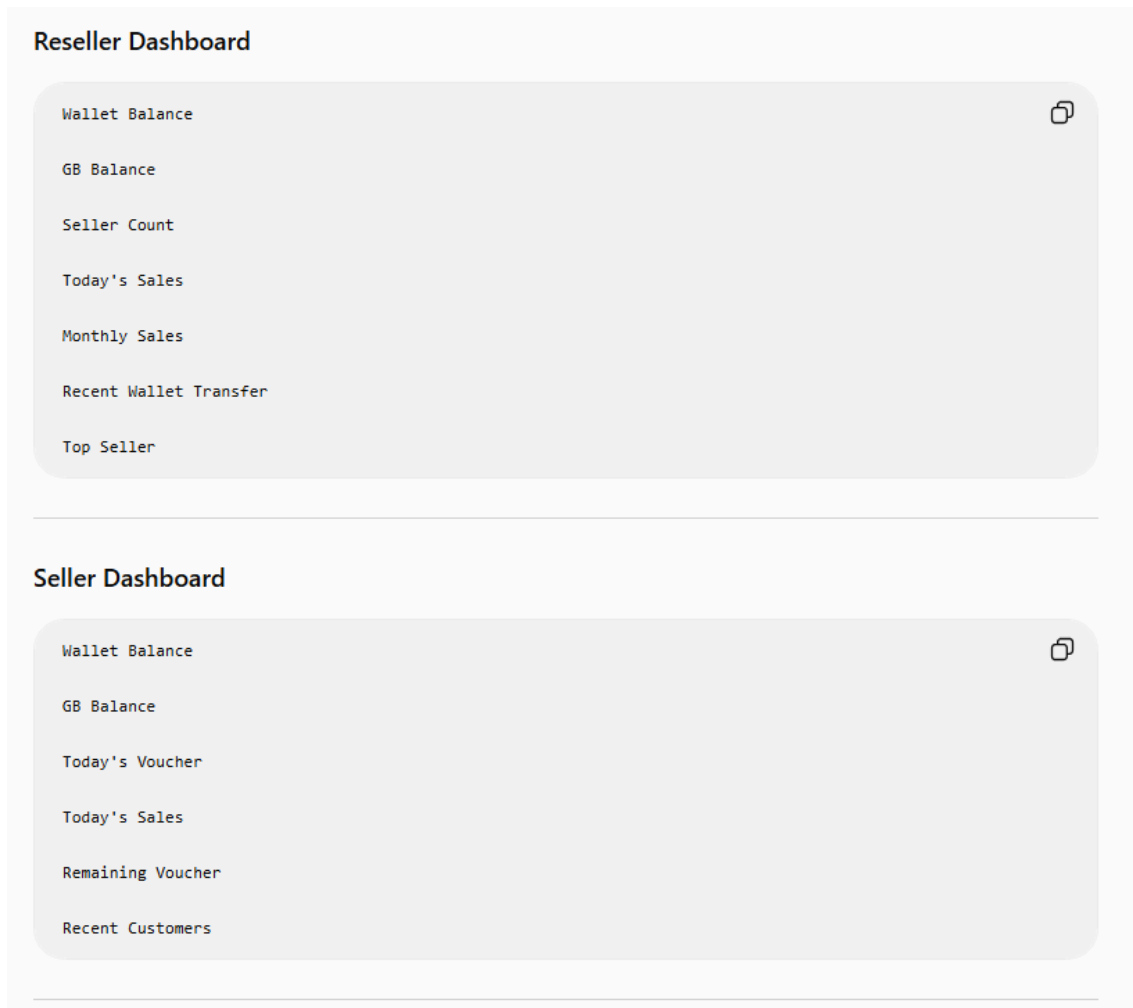


2.A Permission Matrix

Feature	Admin	Reseller	Seller	
Create Plan	✓	✗	✗	
Create Reseller	✓	✗	✗	
Create Seller	✓	✓	✗	
Wallet Load	✓	✓	✗	
Allocate GB	✓	✓	✗	
Generate Voucher	✓	✓	✓	
Delete Voucher	✓	✗	✗	
Reports	✓	✓	Limited	
Dashboard	✓	✓	✓	

2.B Dashboard

Admin Dashboard	
Today's Sales	
Wallet Distributed	
GB Distributed	
Online Users	
Offline Users	
Voucher Generated	
Revenue	
Top Reseller	
Recent Transactions	



3. User Roles

1. Admin

Complete system access.

Permissions

- Create Internet Plans
- Create Resellers
- Create Sellers
- Load Wallet
- Allocate GB

- Set Selling Price
- View Reports
- Disable Users
- Refund Wallet
- View Login Logs
- Manage NAS
- Manage Routers
- Manage FreeRADIUS

2. Reseller

Cannot access system configuration.

Permissions

- Create Seller
- Load Seller Wallet
- Generate Voucher
- View Sales
- View Wallet
- View Seller Report
- Transfer Wallet
- Print Voucher

Restrictions

Cannot create another reseller.

3. Seller

Lowest level.

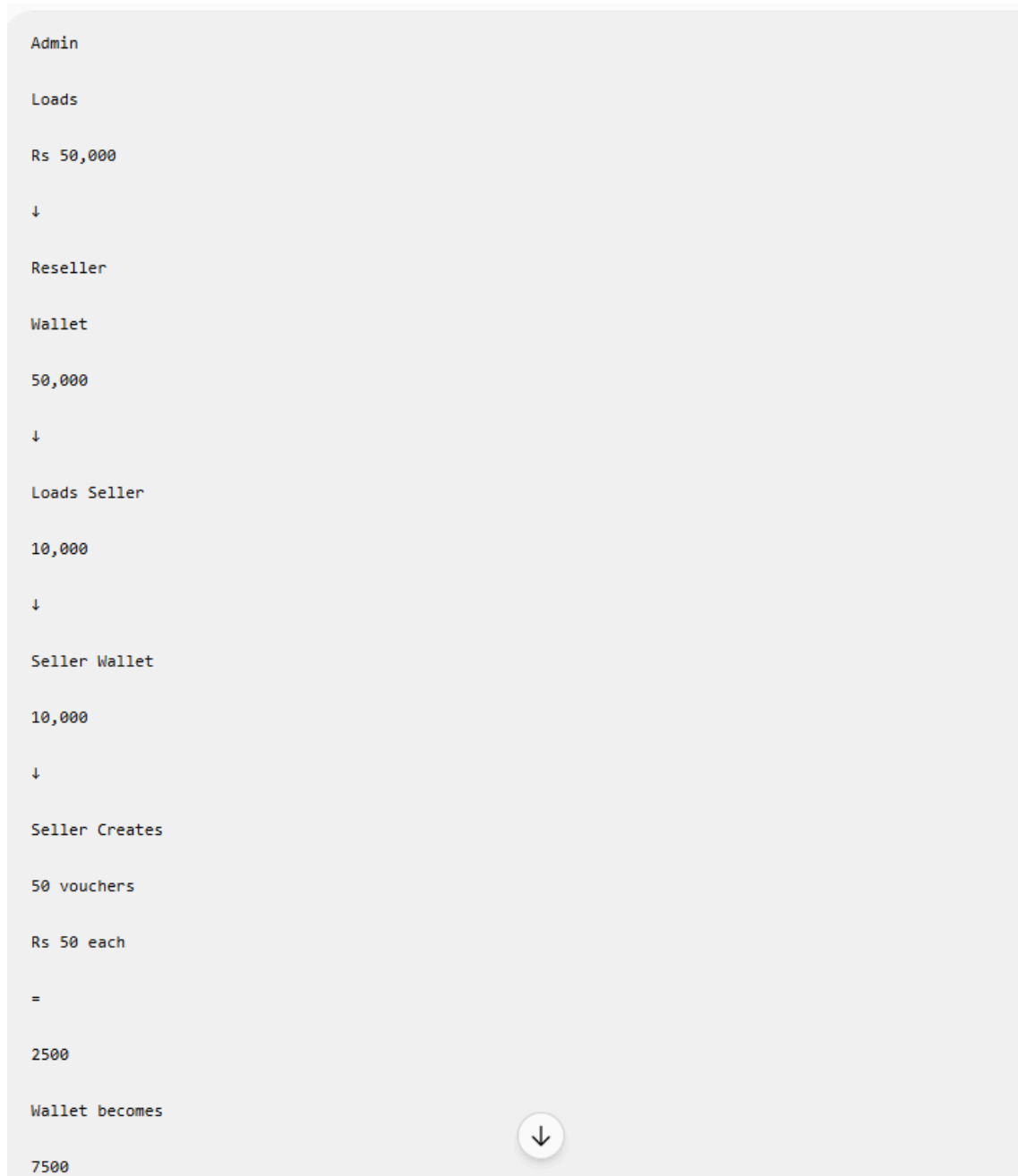
Permissions

- Generate Voucher
- Print Voucher
- View Wallet
- View Sales
- View Customers

Cannot

- Load Wallet
- Create Seller
- Create Plan
- Modify Voucher

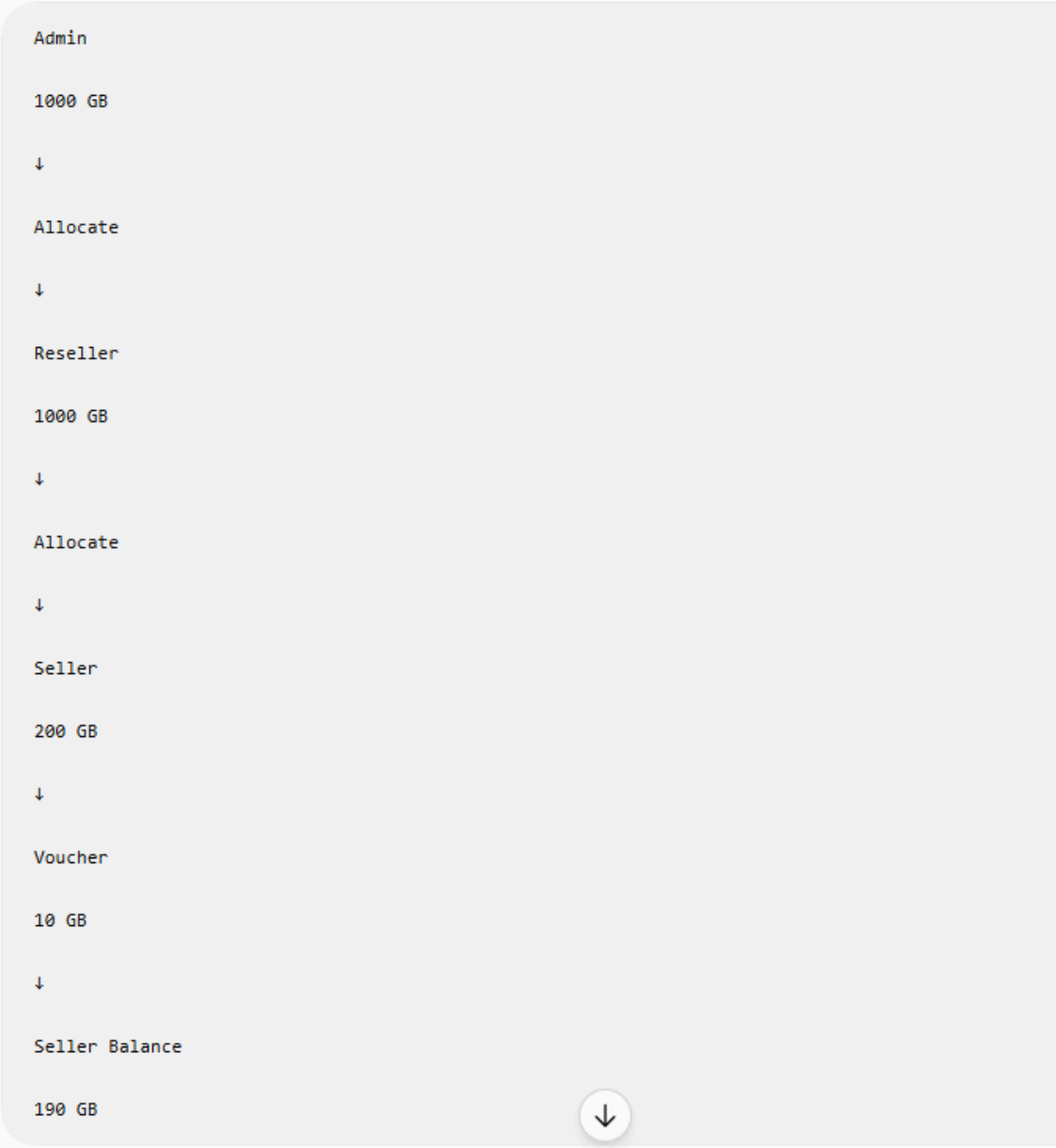
4. Wallet Flow



5. GB Allocation

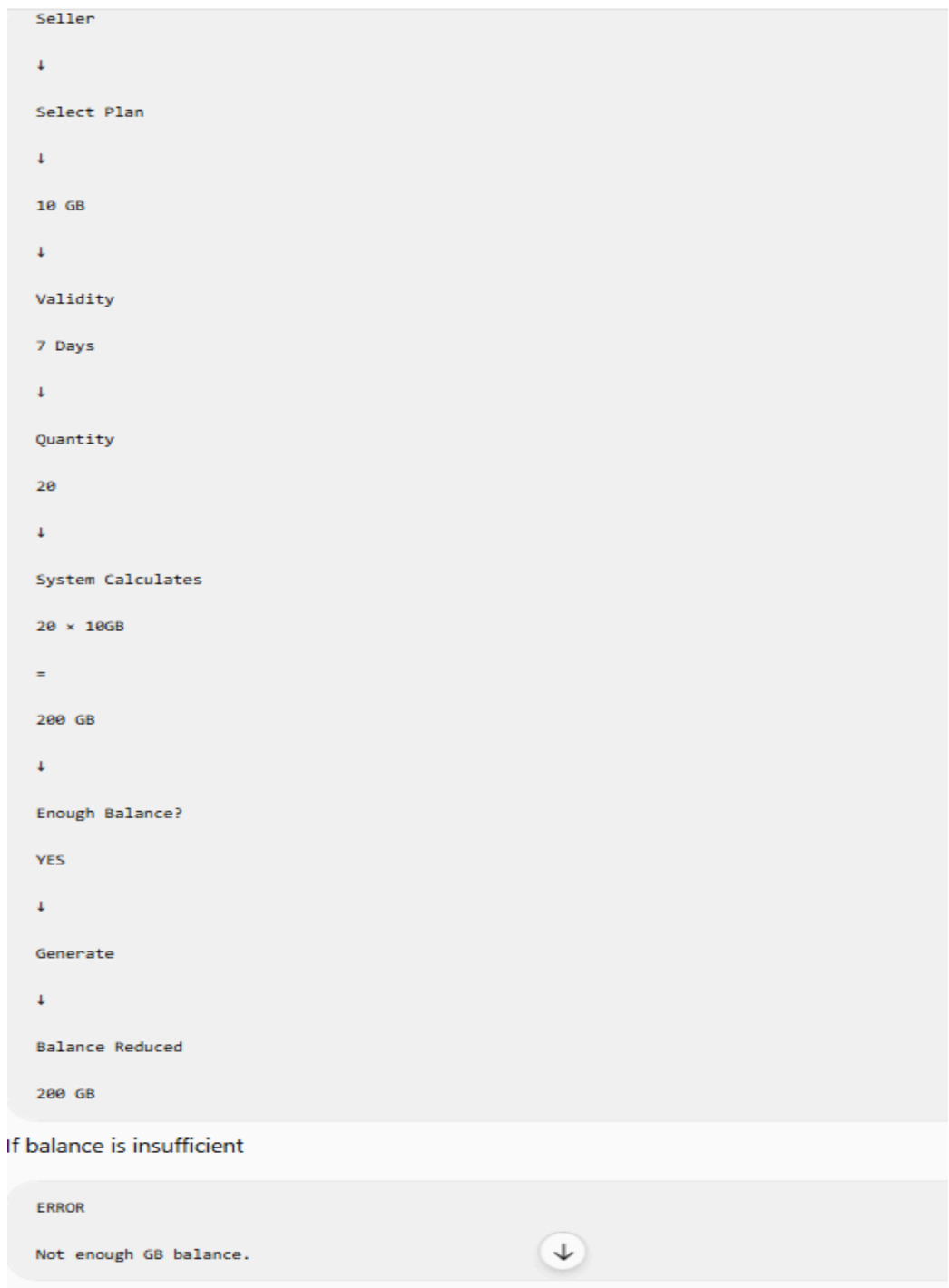
Instead of money only, reseller can also own Internet quota.

Example

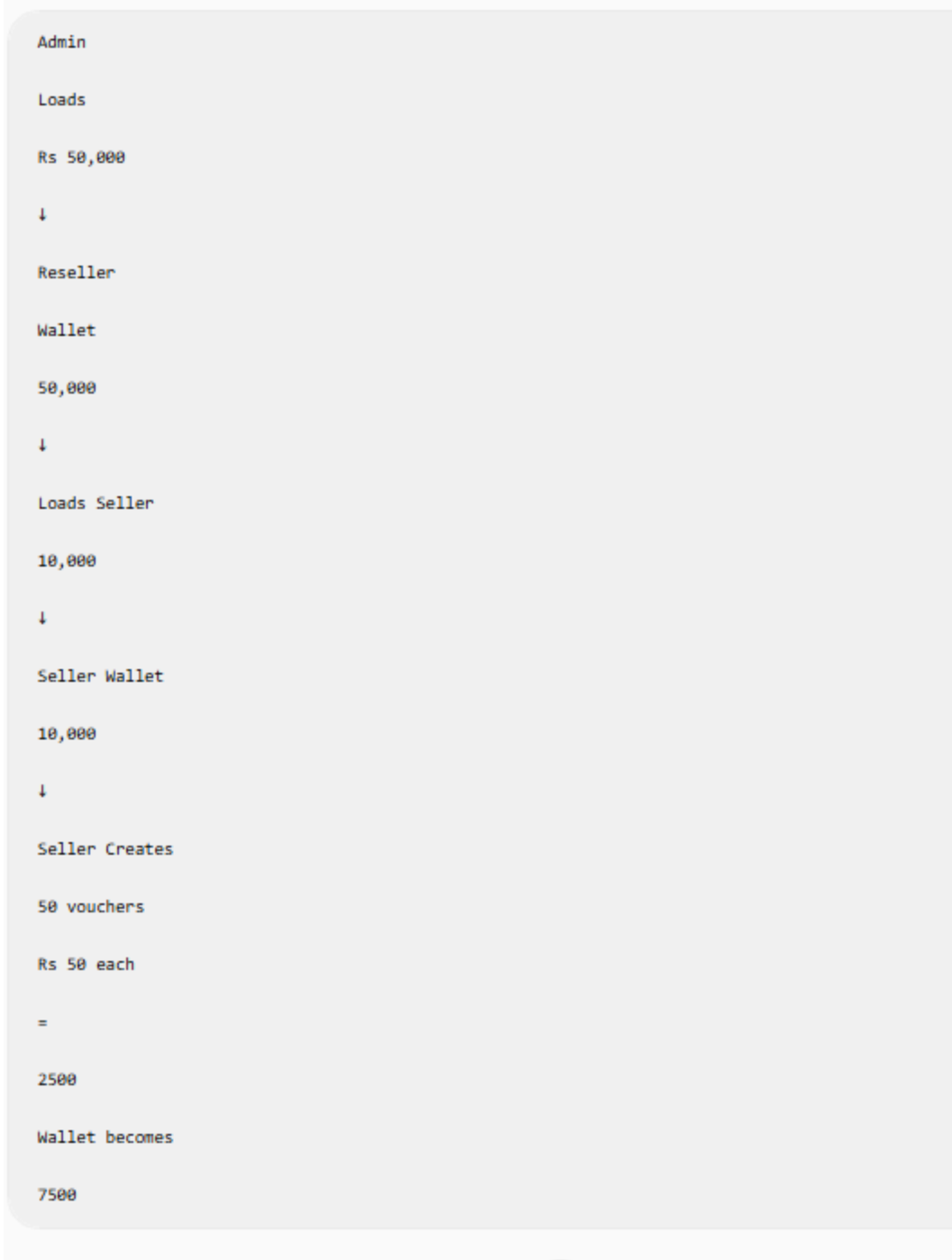


Every generated voucher deducts GB.

6. Voucher Creation Flow



7. Money Flow



8. Voucher Generation Logic

